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GCC3303: Technical Report

**A PREDICTIVE SYSTEM FOR OVARIAN CANCER BASED ON RISK FACTORS USING
BOOSTING TECHNIQUE
BY**

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INTRODUCTION (Background of Study)

- ▶ In recent times, analysis has revealed ovarian cancer (OC) as the seventh most well-known cancer among females, which has a 2.7% lifetime risk factor. Ovarian cancer represents 2.5% of all malignancies among females; however, 5% of the malignant cases die due to low survival rates (Ahamad et al., 2022). Ting-Ting et al., (2021) mentioned that Ovarian Cancer (OC) is the leading cause of mortality among gynecologic malignancies worldwide, although the burden estimates (e.g., incidence, prevalence, and mortality) of this disease show a decrease in recent decades.
- ▶ Moreover, cancer is one of the main diseases that cause death in the world (World Health Organization, 2018). Cancer as a disease in humans often attacks all parts of the body, whereas the attack of a cancer in the ovary, is referred to as “ovarian cancer”. Ovarian cancer ranks fifth as a cause of death in women worldwide (Arfiani & Rustam, 2019; Wu et al., 2020) and kills about 150,000 women per year worldwide (Toh & Brody, 2021 cited Bray et al., 2018). Contrarily, Hsiao et al., (2021) attributed to ovarian cancer (OC), the seventh most common malignancy, which causes the eighth highest mortality rate of all cancer types worldwide.

Statement of the Problem

- ▶ Despite current advances in personalized treatments, Ovarian Cancer still affirms a high iteration rate after adequate treatments, because of its heterogeneous. Due to the limitation of monitoring techniques, there is no early diagnosis for many patients, leading to poor survival. The development of the promising predictive system and risk stratification model are of great importance to helping clinical decision-making and improve survival in the realm of precision medicine for ovarian cancer.
- ▶ At present, the Machine learning (ML) model has been applied to disease screening and diagnosis. To a certain extent, the diagnostic efficiency and the prognosis of patients have been improved. However, in most previous studies, dataset employed seems effective, but it contains many parameters which might affect the efficiency of the predictive system. Therefore, developing predictive models to predict lung metastasis can help guide clinical strategies, which are important for improving the prognosis of patients with ovarian cancer. Therefore, this study aims to adopt a huge dataset with minimum number of parameters to develop an efficient predictive system for ovarian cancer based on risk factors using boosting technique.

Aim and Objectives

- ▶ This research aimed to develop a predictive system for ovarian cancer based on risk factors using boosting technique. In order to achieve this research purpose, the following objectives are put forward.
- ▶ i. Acquisition of dataset from kaggle.com.
- ▶ ii. Dataset preprocessing and feature engineering using variance threshold
- ▶ iii. Implementation of the predictive system using GradientBoostingClassifier.
- ▶ iv. Evaluation of the predictive system using accuracy, precision, recall, and f1-score.

Scope of the Study

- ▶ In this research, a machine learning algorithm will be employed for the proposed ovarian cancer predictive system. Moreover, boosting technique such as Gradient Boosting (GB) technique was proposed in order to empower the accuracy of the proposed predictive system. Furthermore, dataset from Kaggle.com will be used. The reason behind the usage of Kaggle dataset is because of its availability and large dimensions. Finally, the proposed predictive system will be developed using python programming language.

Significance of the study

- The current study is designed to bridge the existing research gaps as regards to early detection/prediction of Ovarian Cancer (OC). Therefore, it aims to utilize boosting techniques for the prediction of Ovarian Cancer (OC). In a nutshell, this research is hopeful to cater for rapid detection of ovarian cancer as immediate detection is required and germane for knowing the vulnerability of this ailment attack in patients. However, there are many techniques that have been carried out and used for ovarian cancer detection and seek for appropriate treatment recommendations. These techniques are found to be the use of computational techniques. In this research, boosting technique is employed to identify whether a patient has ovarian cancer or not.

Gradient Boosting Machine (GBM)

- ▶ GBM according to Ahamad et al., (2022), is an ensemble learning method that merges multiple feeble learners to make a robust one through the optimization of the loss function that normally uses the deviance or exponential loss function. Logistic regression is employed to handle deviance loss function, and Adaboosting is applied to control exponential loss function.
- ▶ In the same vein, Paik et al., (2019) stressed, the gradient boosting (GB) algorithm is one of the most powerful machine learning techniques that has exhibited success in clinical applications. It is one of the ensemble methods that rely on combining many weak simple predictors to obtain a stronger prediction.

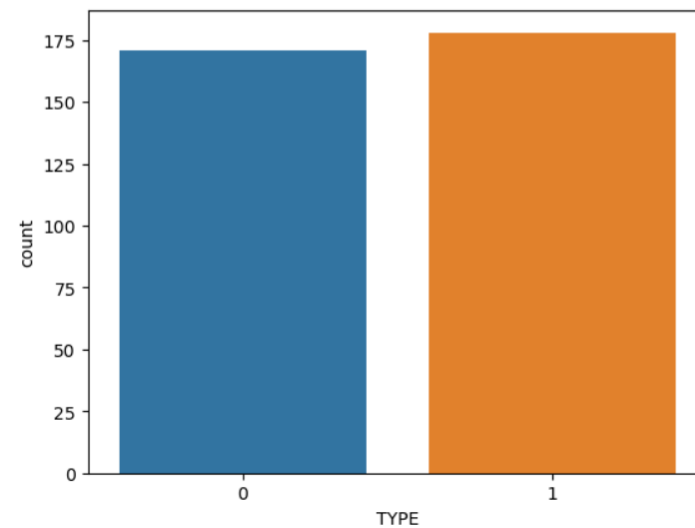
Visualization of Target Column

```
#viewing the distribution of the TYPES column using value_counts  
# 1 = negative, 0 = positive  
ovarian_data['TYPE'].value_counts()
```

```
1    178  
0    171  
Name: TYPE, dtype: int64
```

```
#visualizing the distribution of the TYPES using seaborn library with countplot method  
sns.countplot(x = ovarian_data['TYPE'])
```

```
<Axes: xlabel='TYPE', ylabel='count'>
```



Performance Evaluation

- ▶ Below shows the performance evaluation of the system. It's evaluated using metrics such as precision, accuracy, and f1-score.

- ▶ **Model Evaluation**

Model evaluation is the process of using different evaluation metrics to understand a machine learning model's performance, as well as its strengths and weaknesses.

```
#making prediction with the trained model  
y_pred = gbc_model.predict(x_test)
```

```
#evaluating with classification  
gbc_report = classification_report(y_pred, y_test)  
  
print(gbc_report)
```

	precision	recall	f1-score	support
0	0.94	0.84	0.89	37
1	0.84	0.94	0.89	33
accuracy			0.89	70
macro avg	0.89	0.89	0.89	70
weighted avg	0.89	0.89	0.89	70

```
#evaluating with accuracy_score  
gbc_score = accuracy_score(y_pred, y_test)  
  
print(gbc_score)
```

```
0.8857142857142857
```

System Performance

- ▶ Below shows the weighted average used to for the developed system in this research using Gradient Boosting technique.

MODEL	ACCURACY	PRECISION	RECALL	F1-SCORE
GRADIENT BOOSTING	0.89	0.89	0.89	0.89

Summary

- In this project, a predictive system for ovarian cancer was developed using the gradient boosting technique. The project aimed to leverage machine learning algorithms to accurately predict ovarian cancer cases, enabling early detection and personalized treatment strategies. The project involved various stages, including data acquisition, data preprocessing, model development, and performance evaluation. The dataset used for the project was obtained from Kaggle and underwent rigorous data preprocessing, including handling missing values, feature selection, and normalization. The gradient boosting technique implemented using libraries like GradientBoostingClassifier was employed to develop the predictive model. The model was trained on a training set and evaluated using a testing set to assess its performance based on various classification metrics. Results demonstrated the effectiveness of the developed predictive system in accurately predicting ovarian cancer cases. The model achieved high accuracy, precision, recall, and F1 score, indicating its ability to classify ovarian cancer cases with high accuracy and capture positive cases effectively.

Conclusion

- ▶ In conclusion, the developed predictive system for ovarian cancer using the gradient boosting technique has shown promising results. The system has the potential to aid in the early detection and management of ovarian cancer, which can significantly impact patient outcomes. The accurate predictions and robust performance metrics obtained during evaluation validate the effectiveness of the developed system. The project highlights the importance of data preprocessing, feature selection, and leveraging advanced machine learning techniques for accurate prediction. The combination of appropriate data acquisition, preprocessing techniques, and the gradient boosting algorithm has proven successful in building a reliable predictive model for ovarian cancer.

Recommendations

- ▶ Based on the project's findings, several recommendations can be made for further improvements and future research:
- ▶ i. Further Dataset Exploration: Explore additional datasets or sources to enrich the available data.
- ▶ ii. Model Optimization: Conduct further hyperparameter tuning and optimization to maximize the predictive system's accuracy and generalization. Explore other gradient boosting algorithms, ensemble techniques, or deep learning models to assess their performance in the context of ovarian cancer prediction.
- ▶ iii. External Validation: Validate the developed predictive system using external datasets to assess its performance and generalizability across different populations and settings.
- ▶ iv. Clinical Integration: Collaborate with healthcare professionals and experts in the field to ensure the clinical relevance and applicability of the predictive system. Incorporate domain-specific knowledge and feedback to refine and improve the system's predictive capabilities.

Code Link

- ▶ GitHub: <https://github.com/ayobami4/CPN-Project>

The background features abstract green geometric shapes, including triangles and polygons, in various shades of green, creating a modern and dynamic look.

**▶ THANK YOU
FOR YOUR TIME
AND PATIENCE**